IO_POWER.h File Reference

IO Driver functions for Power IC control. More...

```
#include "IO_Error.h"
```

Macros

Power values

Selects power configuration.

#define	IO_POWER_OFF	0U
#define	IO_POWER_ON	1U
#define	IO_POWER_ON_5V	2U
#define	IO_POWER_ON_6V	3U
#define	IO_POWER_ON_7V	4U
#define	IO_POWER_ON_8V	5U
#define	IO_POWER_ON_9V	6U
#define	IO_POWER_ON_10V	7U

Functions

IO_ErrorType	IO_POWER_Set (ubyte1 pin, ubyte1 mode) Sets a certain mode of the Power IC. More...
IO_ErrorType	IO_POWER_Get (ubyte1 pin, ubyte1 *const state) Returns the current state of a POWER feature. More...
IO_ErrorType	IO_POWER_EnableExtShutOff (ubyte1 group) Configures the external shut off inputs. More...
IO_ErrorType	IO_POWER_GetExtShutOff (ubyte1 group, bool *const state) Reads the state of the external shut off signal. More...

Detailed Description

IO Driver functions for Power IC control.

The Power IC control functions allow to power down the ECU, switching on and off the sensor supply voltages of the ECU, enabling/disabling the power stage enable signal and the safety switches.

The ECU provides 2 separate inputs `IO_K15` and `IO_WAKEUP` for ECU power on and off functionalities.

Note

The ECU does not power down automatically when `IO_K15` and `IO_WAKEUP` is low. When `IO_K15` becomes low, the postrun is still active, and a power down can be now initiated by the application SW.

Remarks

- The ECU shall only be powered down with

  ```
  IO_POWER_Set(IO_K15,
               IO_POWER_OFF)
  ```

  or

  ```
  IO_POWER_Set(IO_WAKEUP,
               IO_POWER_OFF)
  ```

  if `IO_K15` is low in order to ensure a correct shutdown.

- It is the responsibility of the user to check for the right state of `IO_K15` directly before calling `IO_POWER_Set()`.

Attention

- `IO_K15` is level sensitive.
  - If `IO_K15` is low, the ECU will power down completely. Otherwise the ECU's reaction to `IO_POWER_Set(IO_K15, IO_POWER_OFF)` and `IO_POWER_Set(IO_WAKEUP, IO_POWER_OFF)` depends on multiple variables.
  - If `IO_K15` is high, and the resets were enabled in the safety configuration, and the watchdog device error counter has not reached the final threshold value, the ECU will perform a hard reset. The hard reset will reset the entire ECU, including all internal status variables and the watchdog error counter, and then power up the device again. (The current value of the watchdog error counter can be requested with the `DIAG_Status()` function.)
  - In all other cases the shutdown functions have no effect.
- `IO_WAKEUP` is edge sensitive.
  - If `IO_WAKEUP` is high and `IO_POWER_Set()` is called, the ECU will power down and remain off until the next rising edge on `IO_WAKEUP` (assuming that `IO_K15` is low).

Remarks

- Whenever `IO_POWER_Set()` with `IO_POWER_OFF` is called, the postrun will be disabled.
- If the ECU can not power down for some reason (e.g. `IO_K15` is high and a hard reset is not possible) a power down will occur as soon as `IO_K15` goes low.
- The application SW can reenable the postrun by calling `IO_POWER_Set()` with `IO_POWER_ON`.

Power Code Examples

Examples for using the Power API

Example for Power initialization

The Power driver does not need an explicit initialization. The Power driver is initialized by `IO_Driver_Init()`.

```
IO_Driver_Init(NULL);
```

Example for Sensor Supply

```
// switch on sensor supply 0
IO_POWER_Set (IO_SENSOR_SUPPLY_0,
              IO_POWER_ON);

// switch on variable sensor supply 2 with 8 volts
IO_POWER_Set (IO_SENSOR_SUPPLY_2,
              IO_POWER_ON_8V);

// switch off sensor supply 0
IO_POWER_Set (IO_SENSOR_SUPPLY_0,
              IO_POWER_OFF);
```

Example for Safety Switch

```
ubyte1 ssw2_state = IO_POWER_OFF;

// enable safety switch 0 (shut off group 0)
IO_POWER_Set (IO_INT_SAFETY_SW_0,
              IO_POWER_ON);

// get status of safety switch 2 (shut off group 2)
IO_POWER_Get (IO_INT_SAFETY_SW_2,
              &ssw2_state);
```

Macro Definition Documentation

#define IO_POWER_OFF 0U

switch off - turns off the respective power function

#define IO_POWER_ON 1U

switch on - turns on the respective power function

#define IO_POWER_ON_10V 7U

switch on with 10 volts - configures the respective power function for 10 volts and turns it on (only available for `IO_SENSOR_SUPPLY_2`)

#define IO_POWER_ON_5V 2U

switch on with 5 volts - configures the respective power function for 5 volts and turns it on (only available for `IO_SENSOR_SUPPLY_2`)

#define IO_POWER_ON_6V 3U

switch on with 6 volts - configures the respective power function for 6 volts and turns it on (only available for `IO_SENSOR_SUPPLY_2`)

#define IO_POWER_ON_7V 4U

switch on with 7 volts - configures the respective power function for 7 volts and turns it on (only available for `IO_SENSOR_SUPPLY_2`)

#define IO_POWER_ON_8V 5U

switch on with 8 volts - configures the respective power function for 8 volts and turns it on (only available for `IO_SENSOR_SUPPLY_2`)

#define IO_POWER_ON_9V 6U

switch on with 9 volts - configures the respective power function for 9 volts and turns it on (only available for `IO_SENSOR_SUPPLY_2`)

Function Documentation

IO_ErrorType IO_POWER_EnableExtShutOff (ubyte1 group)

Configures the external shut off inputs.

Configures and enables the external shut off inputs for a shut off group

Parameters

group Group for which the external shut off inputs should be configured, one of:

- `IO_INT_SAFETY_SW_0` (shut off group 0)
- `IO_INT_SAFETY_SW_1` (shut off group 1)
- `IO_INT_SAFETY_SW_2` (shut off group 2)

Returns

IO_ErrorType

Return values

IO_E_OK	everything fine
IO_E_INVALID_CHANNEL_ID	the given channel id does not exist or is not available on the used ECU variant
IO_E_CH_CAPABILITY	the given channel is not a valid shut off group
IO_E_CHANNEL_NOT_CONFIGURED	power module has not been initialized
IO_E_CHANNEL_BUSY	the external shut off inputs were already configured or the associated ADC input channels are already used by another function
IO_E_UNKNOWN	an unknown error occurred

Note

The external shut off functionality requires 2 analog inputs for each group. The following pins are used for each group:

- `IO_INT_SAFETY_SW_0` (shut off group 0):
  - `IO_ADC_18` (input 0)
  - `IO_ADC_19` (input 1)
- `IO_INT_SAFETY_SW_1` (shut off group 1):
  - `IO_ADC_20` (input 0)
  - `IO_ADC_21` (input 1)
- `IO_INT_SAFETY_SW_2` (shut off group 2):
  - `IO_ADC_22` (input 0)
  - `IO_ADC_23` (input 1)

Attention

- Both inputs need to switch in opposite direction, using a normally open and a normally close contact.
- The outputs can only be enabled, when input 0 reads a high level and input 1 reads a low level
- The outputs will be disabled, when input 0 reads a low level and input 1 reads a high level

- If both inputs read the same level, the diagnostic state machine will be informed, with the diagnostic error code being **DIAG_E_SSW_EXT_SHUTOFF**.

Remarks

When the external shut off is applied, diagnostic is still possible on the affected outputs

**IO_ErrorType IO_POWER_Get (ubyte1 pin,
ubyte1 *const state
)**

Returns the current state of a POWER feature.

Returns the state of sensor supplies, K15, WakeUp, safety switches and power stage enable

Parameters

pin pin for which the mode shall be set, one of:

- `IO_INT_POWERSTAGE_ENABLE`
- `IO_SENSOR_SUPPLY_0`
- `IO_SENSOR_SUPPLY_1`
- `IO_SENSOR_SUPPLY_2`
- `IO_INT_SAFETY_SW_0` (shut off group 0)
- `IO_INT_SAFETY_SW_1` (shut off group 1)
- `IO_INT_SAFETY_SW_2` (shut off group 2)
- `IO_K15` (K15)
- `IO_WAKEUP` (WakeUp)

[out] **state** pointer to ubyte1, returns the state of the selected feature, one of:

- `IO_POWER_ON`
- `IO_POWER_OFF`
- `IO_POWER_ON_5V`
- `IO_POWER_ON_6V`
- `IO_POWER_ON_7V`
- `IO_POWER_ON_8V`
- `IO_POWER_ON_9V`
- `IO_POWER_ON_10V`

Returns

IO_ErrorType

Return values

IO_E_OK	everything fine
IO_E_INVALID_CHANNEL_ID	the given channel id does not exist or is not available on the used ECU variant
IO_E_CH_CAPABILITY	the given channel is not a POWER channel
IO_E_CHANNEL_NOT_CONFIGURED	power module has not been initialized
IO_E_NULL_POINTER	a NULL pointer has been passed
IO_E_SAFE_STATE	the given channel is in a safe state

Remarks

- IO_INT_POWERSTAGE_ENABLE and IO_INT_SAFETY_SW_0 .. IO_INT_SAFETY_SW_2 are internal pins.
- IO_INT_POWERSTAGE_ENABLE controls the internal powerstage enable signal. Without enabling this signal all power outputs remain low (switched off).
- IO_INT_SAFETY_SW_0 .. IO_INT_SAFETY_SW_2 control the internal safety switches for PWM outputs an represent a shut off group. Without switching this signal to ON, the PWM power stages will not be supplied and the pin will therefore remain low. The safety switch assignment to the output stage is defined as following:
    - IO_INT_SAFETY_SW_0 (shut off group 0): IO_PWM_00 .. IO_PWM_13 (IO_DO_16 .. IO_DO_29)
    - IO_INT_SAFETY_SW_1 (shut off group 1): IO_PWM_14 .. IO_PWM_27 (IO_DO_30 .. IO_DO_43)
    - IO_INT_SAFETY_SW_2 (shut off group 2): IO_PWM_28 .. IO_PWM_35 (IO_DO_44 .. IO_DO_51)
- IO_SENSOR_SUPPLY_0 .. IO_SENSOR_SUPPLY_2 control the 3 external sensor supply pins. IO_SENSOR_SUPPLY_0 and IO_SENSOR_SUPPLY_1 are fixed to 5 volts. Calling this function with this pins, the state can only return IO_POWER_ON of IO_POWER_OFF. IO_SENSOR_SUPPLY_2 is a variable supply and can be configured in the range from 5 to 10 volts. Calling this function with this pin, the state can return IO_POWER_OFF or anything in between IO_POWER_ON_5V and IO_POWER_ON_10V.

```
IO_ErrorType IO_POWER_GetExtShutOff ( ubyte1       group,
                                      bool *const  state
                                    )
```

Reads the state of the external shut off signal.

Reads the evaluated signal of the external shut off inputs

Parameters

	group	Group for which the status shall be read, one of: • `IO_INT_SAFETY_SW_0` (shut off group 0) • `IO_INT_SAFETY_SW_1` (shut off group 1) • `IO_INT_SAFETY_SW_2` (shut off group 2)
[out]	**state**	pointer to bool, returns the state of the shut off group, one of: • `TRUE` if the outputs were disabled by the external shut off • `FALSE` if the outputs were not disabled by the external shut off

Returns

IO_ErrorType

Return values

IO_E_OK	everything fine
IO_E_INVALID_CHANNEL_ID	the given channel id does not exist or is not available on the used ECU variant
IO_E_CHANNEL_NOT_CONFIGURED	the external shut off has not been enabled
IO_E_NULL_POINTER	a NULL pointer has been passed
IO_E_UNKNOWN	an unknown error occurred

IO_ErrorType IO_POWER_Set (ubyte1 pin,
ubyte1 mode
)

Sets a certain mode of the Power IC.

Sets a certain mode for a sensor supply pin or the device power

Parameters

pin pin for which the mode shall be set, one of:

- `IO_INT_POWERSTAGE_ENABLE`
- `IO_SENSOR_SUPPLY_0`
- `IO_SENSOR_SUPPLY_1`
- `IO_SENSOR_SUPPLY_2`
- `IO_INT_SAFETY_SW_0` (shut off group 0)
- `IO_INT_SAFETY_SW_1` (shut off group 1)
- `IO_INT_SAFETY_SW_2` (shut off group 2)
- `IO_K15` (K15, for power down)
- `IO_WAKEUP` (WakeUp, for power down after a wake up)

mode sets a certain mode, one of:

- `IO_POWER_ON`
- `IO_POWER_OFF`
- `IO_POWER_ON_5V`
- `IO_POWER_ON_6V`
- `IO_POWER_ON_7V`
- `IO_POWER_ON_8V`
- `IO_POWER_ON_9V`
- `IO_POWER_ON_10V`

Returns

IO_ErrorType

Return values

IO_E_OK	everything fine
IO_E_INVALID_CHANNEL_ID	the given channel id does not exist or is not available on the used ECU variant
IO_E_CH_CAPABILITY	the given channel is not a POWER channel
IO_E_INVALID_PARAMETER	an invalid parameter has been passed
IO_E_CHANNEL_NOT_CONFIGURED	power module has not been initialized
IO_E_SAFE_STATE	the given channel is in a safe state
IO_E_INTERNAL_CSM	an internal error occurred

IO_E_UNKNOWN	an unknown error occurred

Remarks

- IO_INT_POWERSTAGE_ENABLE and IO_INT_SAFETY_SW_0 .. IO_INT_SAFETY_SW_2 are internal pins.
- IO_INT_POWERSTAGE_ENABLE controls the internal powerstage enable signal. Without enabling this signal all power outputs remain low (switched off).
- IO_INT_SAFETY_SW_0 .. IO_INT_SAFETY_SW_2 control the internal safety switches for PWM outputs and each represents a shut off group. Without switching this signal to ON, the PWM power * stages will not be supplied and the pins will therefore remain low. The safety switch assignment to the output stages is defined as follows:
  - IO_INT_SAFETY_SW_0 (shut off group 0): IO_PWM_00 .. IO_PWM_13 (IO_DO_16 .. IO_DO_29)
  - IO_INT_SAFETY_SW_1 (shut off group 1): IO_PWM_14 .. IO_PWM_27 (IO_DO_30 .. IO_DO_43)
  - IO_INT_SAFETY_SW_2 (shut off group 2): IO_PWM_28 .. IO_PWM_35 (IO_DO_44 .. IO_DO_51)
- IO_SENSOR_SUPPLY_0 .. IO_SENSOR_SUPPLY_2 control the 3 external sensor supply pins. IO_SENSOR_SUPPLY_0 and IO_SENSOR_SUPPLY_1 are fixed to 5 volts. Calling this function with this pins, the mode can only be set to IO_POWER_ON of IO_POWER_OFF. IO_SENSOR_SUPPLY_2 is a variable supply and can be configured in the range from 5 to 10 volts. Calling this function with this pin, the mode can only be set to IO_POWER_OFF or anything in between IO_POWER_ON_5V and IO_POWER_ON_10V.

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